

Typesetting the DPO Loss in L^AT_EX

Daily Toolkit Deep Dive — Section 3

July 23, 2026

1 Inline math

The Direct Preference Optimization (DPO) loss trains the policy π_θ directly on preference pairs (x, y_w, y_l) , where y_w is the preferred (“chosen”) response and y_l is the dispreferred (“rejected”) response, without ever training a separate reward model. The temperature $\beta > 0$ controls how far π_θ is allowed to drift from the frozen reference policy π_{ref} .

2 Display math: the DPO loss

$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_\theta(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right) \right] \quad (1)$$

Equation (1) is exactly the quantity implemented by `dpo_loss` in `dpo_minimal.py`: the two log-ratio terms inside $\sigma(\cdot)$ correspond to `log_ratio_w` and `log_ratio_l`, and the whole bracketed expression before the leading minus sign and expectation corresponds to `margin`, which is passed through `F.logsigmoid`.

3 A multi-line derivation with align

The two implicit rewards used inside Equation (1) can be written out separately using the `align` environment from `amsmath`, which lines up the equals signs across rows:

$$\hat{r}_w = \beta \log \frac{\pi_\theta(y_w | x)}{\pi_{\text{ref}}(y_w | x)} \quad (2)$$

$$\hat{r}_l = \beta \log \frac{\pi_\theta(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \quad (3)$$

so that Equation (1) can be read compactly as $\mathcal{L}_{\text{DPO}} = -\mathbb{E}[\log \sigma(\hat{r}_w - \hat{r}_l)]$.